AP CALCULUS AB

Unit 1 • Worksheet 1-7A • Answer key

Worksheet 1-7A • Instantaneous Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{h\to 0} \dfrac{(3+h)^2 - 9}{h}$ is

(A) 9

(B) 0

(C) 6

(D) 3

Solution

Answer: (C). It is $f'(3)$ for $f(x) = x^2$.

Question 2

NO CALCULATOR

$\lim_{h\to 0} \dfrac{\sqrt{4+h} - 2}{h}$ represents $f'(a)$ for which f and a?

(A) $f(x) = \sqrt{x}$, $a = 4$

(B) $f(x) = \sqrt{x}$, $a = 2$

(C) $f(x) = \sqrt{x+4}$, $a = 2$

(D) $f(x) = x^2$, $a = 4$

Solution

Answer: (A).

Question 3

NO CALCULATOR

Using $g'(x) = 3x^2$ for $g(x) = x^3$, which is the tangent line to $y = x^3$ at $x = 1$?

(A) $y = 3x - 2$

(B) $y = 3x + 1$

(C) $y = x$

(D) $y = 3x$

Solution

Answer: (A). Point $(1, 1)$, slope 3: $y - 1 = 3(x - 1)$.

Question 4

NO CALCULATOR

At which point is $f(x) = |x - 2| + 1$ continuous but not differentiable?

(A) $x = 1$
(B) $x = 2$
(C) $x = -2$
(D) nowhere

Solution
Answer: (B). Corner.

Question 5

NO CALCULATOR

s is differentiable; $s(2) = 9$, $s(4) = 20$, $s(6) = 35$. Which is the best estimate of $s'(4)$?

(A) 5.5
(B) 6.5
(C) 7.5
(D) 20

Solution
Answer: (B). $\frac{35-9}{4}$.

Question 6 (Free response)

NO CALCULATOR

(a) Use the definition to find $f'(x)$ for $f(x) = x^2 - 3x$.
(b) Write the tangent line at $x = 4$.
(c) Is $g(x) = \begin{cases} x^2, & x \le 1 \\ 2x - 1, & x > 1 \end{cases}$ differentiable at $x = 1$? Explain.

Solution
(a) $\frac{(x+h)^2-3(x+h)-x^2+3x}{h} = 2x + h - 3 \to 2x - 3$.
(b) $f(4) = 4$, $f'(4) = 5$: $y - 4 = 5(x - 4)$.
(c) Continuous at 1 (both give 1); slopes 2 and 2 match: yes, $g'(1) = 2$.